Nora

A Gemma-powered companion for tracking, understanding, and speaking up about menstrual and pelvic health

Project writeup — Kaggle: Build with Gemma Comeback Sprint

1. Overview

Nora is a menstrual and pelvic-health companion app built around a simple premise: most cycle trackers ask users to log data, but very few help users make sense of it, and almost none help users translate months of symptoms into something a clinician can act on. Bloom closes that gap by pairing low-friction, high-signal logging with an on-device Gemma model, named Luna within the product , that narrates patterns back to the user in plain language, and by automatically compiling a clinical-grade summary a user can bring to an appointment.

The product is designed first for people managing conditions with a cycle-linked signature that is hard to describe to a doctor in a ten-minute visit ; endometriosis, PCOS, PMDD, and adenomyosis are the leading examples. Diagnostic delay for endometriosis alone averages 7-10 years globally, largely because symptom patterns are never captured in a form a clinician can use. Nora's core bet is that a structured, longitudinal symptom record, produced with almost no logging effort, is more valuable than another prediction calendar.

2. Core experience

The app is organized around three screens: Today (daily logging and check-ins), the 3D Symptom Mapper (detailed pain logging), and My Health Story (the compiled clinical summary). A persistent cycle calendar and phase indicator (e.g. "Follicular, day 9") sits at the top of every screen, alongside a one-tap "SOS Flare" control for logging an acute symptom spike without navigating away from whatever the user is doing.

2.1 Today — daily check-in

The home screen leads with Luna, a calm, mood-reactive avatar (rendered here as a dew-drop) whose expression and copy shift with the user's logged energy level, keeping the daily check-in from feeling like a form. A single slider captures a "3-second check-in" — an energy rating from low to radiant — and a row of quick-symptom chips (Bad Cramps, Endo Belly, Radiating Leg Pain, Heavy Flow, Nausea) lets common symptoms be logged in one tap. A horizontally scrolling cycle calendar shows the current day in context without requiring the user to open a separate calendar view.

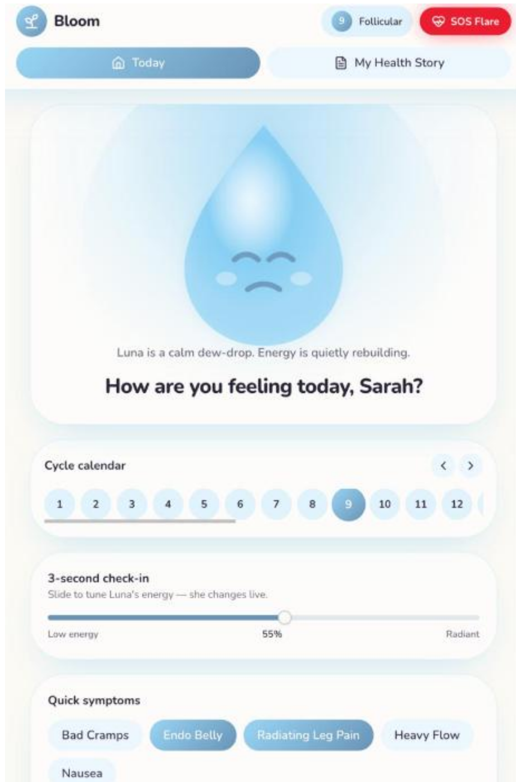

Fig. 1 — Today screen: Luna, the 3-second check-in slider, and quick-symptom chips.

2.2 3D Symptom Mapper — precise, low-friction pain logging

Pain location and character are often the most clinically useful signal and the hardest to capture through a text box. The Symptom Mapper renders a rotating anatomical wireframe; the user taps a region to drop a glowing pain point, then adjusts intensity (a 1-10 slider) and depth (surface / mid / deep) for that point. Multiple points can be mapped per entry — the example below shows five points concentrated around the lower abdomen and pelvis, exactly the kind of clustered, radiating pattern that is difficult to convey verbally but obvious once visualized.

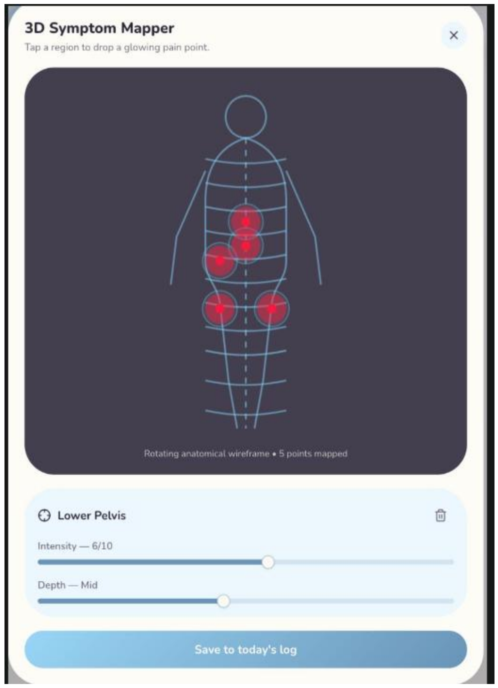

Fig. 2 — 3D Symptom Mapper: five pain points logged around the lower pelvis, with intensity and depth controls.

2.3 Talk to Luna ; the Gemma layer

Talk to Luna is the conversational surface for the underlying Gemma model, which runs locally on the user's device rather than calling out to a cloud API. The interface makes this explicit ("Powered by Gemma running on your machine — asks go straight to the local model"), which matters for a product handling sensitive reproductive-health data: nothing about a conversation needs to leave the device to get an answer.

Luna answers using the user's own logged history as context, so a question like "Why am I so tired today?" is answered with reference to that day's logged symptoms and recent cycle activity rather than a generic response. Responses are deliberately framed as supportive and explanatory, never diagnostic — Luna narrates what the data shows and encourages rest, tracking, or a clinical follow-up; it does not tell the user what condition they have.

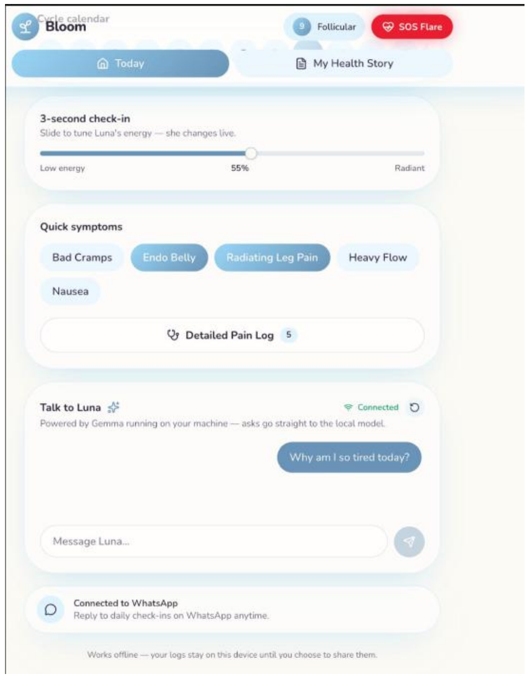

Fig. 3 — Talk to Luna: an on-device Gemma chat grounded in the user's own log data, with a WhatsApp bridge for check-ins outside the app.

For users who don't want to open the app daily, the same check-in flow is available through a WhatsApp bridge, so a quick symptom log can be sent as an ordinary text message and still land in the same on-device store.

2.4 My Health Story — the clinical bridge

This is Nora's central value proposition made visible: a one-page summary, automatically compiled from months of logs and written in the language clinicians use, so a patient is heard the first time rather than having to reconstruct months of symptoms from memory in a rushed appointment. The example below aggregates three months of data into headline metrics days missed from work or school, NSAID efficacy, average peak pain, and cycles logged followed by a cycle-by-cycle summary showing flow, duration, and the standout symptom for each month (e.g. "Heavy (6 days) — Radiating leg pain, nausea").

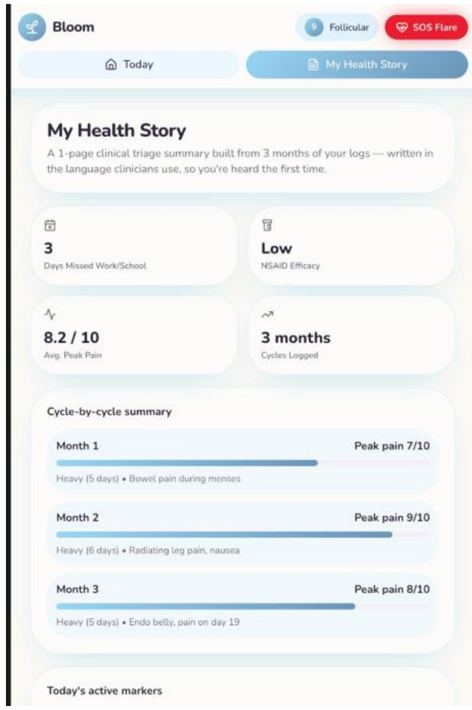

Fig. 4 — My Health Story: a clinician-readable summary compiled automatically from three months of logs.

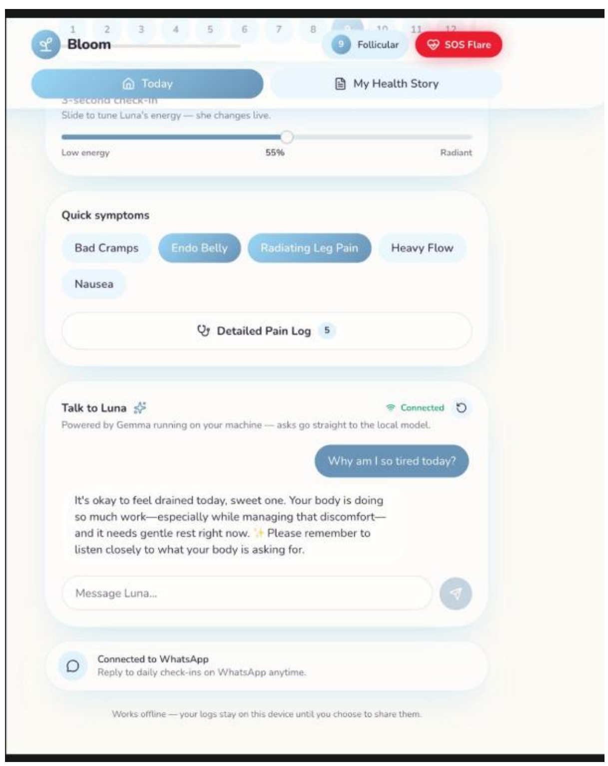

Fig. 5 — Full Today screen scroll, showing the check-in slider, quick symptoms, Talk to Luna, and the WhatsApp bridge together.

3. Why Gemma, and why on-device

- Privacy by architecture, not just policy: menstrual, pain, and fertility data is some of the most sensitive personal data a person generates. Running Gemma locally means a user's symptom history and conversations with Luna never have to leave their device to produce an answer.
- Offline-first for real-world use: logging and chat continue to work without connectivity, which matters for users in low-connectivity settings or who simply don't want a health app phoning home constantly.
- Grounded, not generative-only: Gemma's role is to narrate patterns that a deterministic layer has already computed from the user's structured logs, rather than freely inferring medical claims — the model explains and contextualizes, it does not diagnose.
- Cost and latency: on-device inference avoids per-conversation API costs and network round trips, which is what makes a same-day, always-available "ask Luna anything" chat viable as a daily habit rather than an occasional feature.

4. Design principles

- Logging must be faster than not logging. Sliders, chips, and one-tap SOS entries exist because every additional form field is a point where users stop being honest or stop logging altogether.
- Pattern-flagging, not diagnosis. Bloom surfaces what the data shows and encourages a clinical conversation; it deliberately avoids naming conditions or making diagnostic claims.
- Value has to show up early. A three-month clinical summary is the anchor feature, but the app front-loads smaller, immediate payoffs — Luna's reactive check-ins, same-day symptom insight — so users don't have to wait months to feel the product is working.
- Retention through utility, not dark patterns. There are no streaks, guilt notifications, or manufactured urgency anywhere in the flow. The repeat-use driver is that the data becomes more useful the longer it's tracked, and that the health story is something users genuinely need for real appointments.

5. Feature summary

Screen / feature	What it does
Today home screen	Live cycle-day tracker, a mood-reactive companion avatar ("Luna"), a 3-second energy check-in slider, and one-tap quick-symptom chips.
3D Symptom Mapper	A rotating anatomical wireframe where users drop glowing pain points onto the body, then set intensity and depth per point — richer signal than a text field, at near-zero typing cost.
Talk to Luna	An on-device chat companion, powered by Gemma running locally on the user's machine, that answers questions in natural language using the user's own logged data as context.
My Health Story	An auto-generated, clinician-readable one-page summary compiled from months of logs: missed school/work days, NSAID efficacy, average peak pain, and a cycle-by-cycle timeline.

Screen / feature	What it does
WhatsApp bridge	Daily check-ins can be answered as ordinary WhatsApp messages for users who don't want to open the app.

6. Target users and conditions

Bloom is built primarily for people tracking symptoms tied to conditions with a cycle-linked signature that's easy to dismiss without structured data:

- Endometriosis — pain disproportionate to flow, pain before/after menses, GI symptoms, radiating pain.
- PCOS — irregular or absent cycles, longer cycle length, cross-symptom correlations (weight, acne, hair growth).
- PMDD — severe mood symptoms tightly clustered in the luteal phase.
- Adenomyosis and fibroids — heavy, prolonged bleeding and clotting patterns.

For all of these, the mapper and quick-symptom logging build the raw record, and My Health Story turns that record into the artifact that actually changes a clinical conversation.

7. Status and next steps

The screens shown here represent the current interactive prototype covering the primary logging, chat, and clinical-summary flows. Planned next steps include: expanding the rule-based pattern-flagging layer that feeds Luna's context (cycle-length variance, luteal-phase symptom clustering, GI correlation) so its explanations are grounded in explicit, auditable flags rather than the raw log alone; adding wearable-sourced passive signals (temperature, sleep, heart rate) to reduce manual logging further; and validating the flagging thresholds with clinical input before any suggested pattern is shown as anything more than an early signal.